P02

Multicast topology inference from measured end-to-end loss

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Domain Networking & distributed systems

Abstract

The use of multicast inference on end-to-end measurement has recently been proposed as a means to infer network internal characteristics such as packet link loss rate and delay. In this paper, we propose three types of algorithm that use loss measurements to infer the underlying multicast topology: i) a grouping estimator that exploits the monotonicity of loss rates with increasing path length; ii) a maximum-likelihood (ML) estimator (MLE); and iii) a Bayesian estimator. We establish their consistency, compare their complexity and accuracy, and analyze the modes of failure and their asymptotic probabilities.

Real-World Impact & Application

Finding Where Your Wi-Fi or Game is Lagging: When your multiplayer game (Valorant, Fortnite) spikes to 300ms ping or your stream stutters, you usually cannot see which router on the internet is dropping packets. This paper showed how sending test packets to multiple friends lets you pinpoint the exact lagging router from the outside.